

Data Collection Plan

For our MVP (Minimum Viable Product), we started with a reliable and ready-made dataset:

the Apple Leaf Disease dataset from PlantVillage on Kaggle.

This dataset is widely used in research, well-organized, and contains thousands of labeled images of apple leaf diseases. It allowed us to quickly train a working YOLO classification model and build the full pipeline end-to-end (image upload → prediction → AI advice) without waiting for custom data collection.

However, this dataset only covers **one type of crop**, and our long-term goal is much bigger—we want to detect diseases in **ornamental plants**, the plants found in homes, gardens, and nurseries. This is where we faced our biggest challenge: **ornamental plant data is extremely scarce.**

Unlike crops, ornamental plants have:

- Very few organized datasets online
- Extremely high diversity (hundreds of varieties)
- Many species with similar-looking leaves
- Poor documentation of diseases
- Very few labeled images available publicly

Because of this, we cannot rely on Kaggle or PlantVillage for ornamental plant data.

So our approach is to **build our own dataset manually**, step by step.

Our Plan for Manual Data Collection

1. Working With Experts

We will collaborate with horticulture experts, nursery owners, and plant enthusiasts.

They will help us:

- Identify plant species correctly
- Confirm visible disease symptoms
- Avoid mislabeling similar-looking plants

This expert involvement ensures accuracy.

2. Taking Our Own Photographs

Since ornamental plant images are not available in large numbers, we will take photos ourselves by visiting:

- Nurseries
- Public parks
- Home gardens
- Terrace gardens

We will capture close-ups of leaves, flowers, stems, and any visible infection. This gives us original, high-quality, and copyright-free data.

3. Using Books and Botanical References

We will refer to:

- Gardening guidebooks
- Floriculture manuals
- Horticulture textbooks
- Online botanical databases

These sources help us verify plant names, varieties, and disease descriptions, even if we can't use their images directly.

4. Collecting Real Disease Cases

Most ornamental plant disease images online are either rare or low quality.

So we plan to document real-world disease cases ourselves whenever we come across them.

This includes:

- Early infection
- Fully spread infection
- Environmental stress symptoms
- Multiple angles of the same leaf

It makes the dataset more realistic.

5. Expanding Slowly and Consistently

We will build the dataset gradually, by continuously adding:

- New species
- New diseases
- Seasonal samples
- Better quality images

This slow expansion ensures steady improvement without compromising quality.

6. Verification and Clean Labeling

Every image we collect will be:

1. First labeled by us
2. Cross-checked with reference materials
3. Confirmed with experts whenever needed

This keeps the dataset clean and trustworthy.

